

C882

Geometry for Secondary Mathematics Teaching

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

In a unit that a teacher built around dynamic geometry software and patty-paper tasks, students map triangle ABC to triangle A'B'C' with a 90° rotation about the origin followed by a dilation centered at the origin with scale factor 2. The class meets in a double period on Thursdays. Which statement about the two triangles is correct?

- A. They are congruent, because a rotation is a rigid motion.
- B. They are neither, because a sequence of two different transformations never preserves shape.
- C. They are similar but not congruent; angles match and each side of A'B'C' is twice as long.
- D. They are similar, and each side of A'B'C' is four times as long.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A teacher reviewing work notes that a student who usually writes very neat two-column proofs showed $AB = DE$, $BC = EF$, and $\angle A \cong \angle D$ in triangles ABC and DEF, then concluded the triangles are congruent. The proof was turned in two days late. Which response is mathematically correct?

- A. Not justified; two sides and a non-included angle (SSA) do not guarantee congruence.
- B. Not justified, because late work must be rewritten before it can be judged.
- C. Justified by SAS.
- D. Justified, because any three pairs of congruent parts prove congruence.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

In triangles PQR and XYZ, $PQ = XY$ and $QR = YZ$. Which one additional fact is enough to prove the triangles congruent?

- A. $\angle P \cong \angle X$
- B. $\angle Q \cong \angle Y$
- C. The two triangles face the same direction on the page.
- D. $\angle R \cong \angle Z$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

At 3 p.m. on a clear day, a flagpole that the parent association installed in 2009 casts a 24-ft shadow. At the same moment a 5-ft post casts a 4-ft shadow. The temperature was 78°F. How tall is the flagpole?

- A. 30 ft
- B. 24 ft
- C. 19.2 ft
- D. 25 ft

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A teacher shows a task on the class’s interactive whiteboard: line m passes through (1, 2) and (5, 4), and it was designed to mirror a ramp in the school’s new wing. What is the slope of a line perpendicular to m?

- A. 2
- B. $-1/2$
- C. $1/2$
- D. -2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Line k has slope $-3/4$. A line through the origin is perpendicular to k. Which point lies on that line?

- A. (4, -3)
- B. (-3 , 4)
- C. (3, 4)
- D. (4, 3)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In a lesson that compares a cone and a cylinder by pouring rice, a teacher uses a cone with radius 3 cm and height 10 cm; its slant height is about 10.4 cm. The rice cost \$2 a bag. What is the cone's volume?

- A. $30\pi \text{ cm}^3$ (about 94.2 cm^3)
- B. $90\pi \text{ cm}^3$
- C. $10\pi \text{ cm}^3$
- D. $31.2\pi \text{ cm}^3$

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

A cone and a cylinder have the same base radius and the same volume. How does the cone's height compare with the cylinder's height?

- A. The cone is 1.5 times as tall.
- B. They are the same height.
- C. The cone is one-third as tall.
- D. The cone is three times as tall.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9.

A teacher using graphing software projects $x^2 + y^2 - 6x + 4y - 12 = 0$ and notes that Greek geometers studied conics more than 2,000 years ago. What are the center and radius of this circle?

- A. Center (3, -2), radius $\sqrt{12}$
- B. Center (3, -2), radius 5
- C. Center (-3, 2), radius 5
- D. Center (3, -2), radius 25

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

A ramp for the school garden rises at 30° to the ground; its sloped surface is 12 ft long, and it was built by a volunteer crew over two weekends. How high above the ground is the top of the ramp?

- A. 24 ft
- B. About 6.9 ft
- C. 6 ft
- D. About 10.4 ft

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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11. [TRANSFER — reversed stem / different variable]

A 10-ft ladder leans against a wall with its foot 5 ft from the wall. What angle does the ladder make with the ground?

- A. About 26.6°
- B. About 63.4°
- C. 30°
- D. 60°

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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12.

In a classroom with new tablets and a geometry app the district bought last spring, a student sorts shapes and says a figure cannot be a rectangle “because it looks like a square.” The student is a strong reader and enjoys drawing. Which next step best fits the student’s level of thinking?

- A. Give extra drawing tasks, since the student enjoys drawing.
- B. Have the student measure sides and angles of squares and rectangles, list the properties, and compare the lists.
- C. State the definition once; the student’s grade level means they will always reason formally.
- D. Assign a formal proof that every square is a rectangle.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

In a unit ~~that a teacher built around dynamic geometry software and patty paper tasks~~, students map triangle ABC to triangle A'B'C' with a 90° rotation about the origin followed by a dilation centered at the origin with scale factor 2. ~~The class meets in a double period on Thursdays.~~ Which statement about the two triangles is correct?

Target word: congruent or similar after the sequence **Decides:** The dilation with $k = 2$ scales lengths; angles stay the same.

A	Dropped element	Ignores the dilation that follows the rotation.
B	Absolute language	Rotations and dilations both preserve angle measure, so shape is kept.
C	Correct	The rotation preserves everything; the dilation keeps angles and doubles lengths.
D	Confusion set	4 is the area factor (k^2), not the length factor.

2. Answer A Concept 2

A teacher reviewing work notes that a student ~~who usually writes very neat two-column proofs~~ showed $AB = DE$, $BC = EF$, and $\angle A \cong \angle D$ in triangles ABC and DEF, then concluded the triangles are congruent. ~~The proof was turned in two days late.~~ Which response is mathematically correct?

Target word: is the congruence justified **Decides:** $\angle A$ is not between AB and BC, so this is SSA.

A	Correct	The included angle for AB and BC is $\angle B$; SSA can produce two different triangles.
B	Camouflage	Answers the lateness detail, not the mathematics.
C	Confusion set	SAS needs the angle between the two sides; $\angle A$ is not.
D	Absolute language	SSA and AAA are three pairs that do not.

3. Answer B Concept 2 · transfer

In triangles PQR and XYZ, $PQ = XY$ and $QR = YZ$. Which one additional fact is enough to prove the triangles congruent?

Target word: which added fact completes a criterion **Decides:** The included angle is $\angle Q$ (and $\angle Y$).

A	Confusion set	$\angle P$ is not between PQ and QR; that is SSA.
B	Correct	The angle between the two known sides gives SAS.
C	Adjacent category	Orientation is not a congruence fact; reflections still give congruent figures.
D	Dropped element	Also a non-included angle; the “included” condition is dropped.

4. Answer A Concept 3

At 3 p.m. on a clear day, a flagpole ~~that the parent association installed in 2009~~ casts a 24-ft shadow. At the same moment a 5-ft post casts a 4-ft shadow. ~~The temperature was 78°F.~~ How tall is the flagpole?

Target word: flagpole height **Decides:** Height/shadow = 5/4 for both objects at the same moment.

A	Correct	$5/4 \times 24 = 30$ ft.
B	Wrong stem number	Reports the shadow length, not the height.
C	Inverted operation	Uses 4/5 instead of 5/4; the ratio is flipped.
D	Confusion set	Additive thinking: the post is 1 ft taller than its shadow, so $24 + 1$.

5. Answer D Concept 4

A teacher shows a task ~~on the class's interactive whiteboard~~: line m passes through $(1, 2)$ and $(5, 4)$, ~~and it was designed to mirror a ramp in the school's new wing~~. What is the slope of a line perpendicular to m ?

Target word: perpendicular slope **Decides:** Slope of $m = 2/4 = 1/2$.

A	Inverted operation	Took the reciprocal but did not change the sign.
B	Dropped element	Changed the sign but did not take the reciprocal.
C	Confusion set	That is the parallel slope.
D	Correct	Negative reciprocal of $1/2$.

6. Answer C Concept 4 • transfer

Line k has slope $-3/4$. A line through the origin is perpendicular to k . Which point lies on that line?

Target word: a point on the perpendicular line **Decides:** Perpendicular slope $= 4/3$.

A	Confusion set	Slope $-3/4$: parallel to k , not perpendicular.
B	Inverted operation	Slope $-4/3$: reciprocal taken, sign not changed.
C	Correct	Slope from the origin is $4/3$.
D	Dropped element	Slope $3/4$: sign changed, reciprocal skipped.

7. Answer A Concept 5

In a lesson that compares a cone and a cylinder by pouring rice, a teacher uses a cone with radius 3 cm and height 10 cm; ~~its slant height is about 10.4 cm. The rice cost \$2 a bag~~. What is the cone's volume?

Target word: cone volume **Decides:** $V = (1/3)\pi r^2 h$ with the perpendicular height 10.

A	Correct	$(1/3)\pi(9)(10) = 30\pi$.
B	Dropped element	Forgot the $1/3$; that is the cylinder.
C	Confusion set	Used r instead of r^2 .
D	Camouflage	Used the slant height 10.4 instead of the height.

8. Answer D Concept 5 • transfer

A cone and a cylinder have the same base radius and the same volume. How does the cone's height compare with the cylinder's height?

Target word: height ratio for equal volume **Decides:** $(1/3)Bh_{\text{cone}} = Bh_{\text{cyl}}$, so $h_{\text{cone}} = 3h_{\text{cyl}}$.

A	Adjacent category	Borrows the sphere-to-cylinder $2/3$ ratio, a different solid.
B	Dropped element	Ignores the $1/3$ in the cone formula.
C	Inverted operation	Applies the $1/3$ in the wrong direction.
D	Correct	The $1/3$ factor must be made up by tripling the height.

9. Answer B Concept 6

A teacher ~~using graphing software projects $x^2 + y^2 - 6x + 4y - 12 = 0$ and notes that Greek geometers studied conics more than 2,000 years ago.~~ What are the center and radius of this circle?

Target word: center and radius **Decides:** $(x - 3)^2 + (y + 2)^2 = 12 + 9 + 4 = 25$.

A	Wrong stem number	Used the constant 12 without adding the completed-square terms.
B	Correct	Completing the square gives $r^2 = 25$.
C	Inverted operation	Center signs flipped; $(x - h)$ means $h = +3$.
D	Dropped element	Reported r^2 as the radius.

10. Answer C Concept 7

A ramp for the school garden rises at 30° to the ground; its sloped surface is 12 ft long, ~~and it was built by a volunteer crew over two weekends.~~ How high above the ground is the top of the ramp?

Target word: vertical rise **Decides:** Rise is opposite 30° and the 12 ft is the hypotenuse, so sine.

A	Inverted operation	$12 \div \sin 30^\circ$ divides instead of multiplying.
B	Adjacent category	$12 \times \tan 30^\circ$ treats the ramp as the adjacent leg.
C	Correct	$12 \times \sin 30^\circ = 6$.
D	Confusion set	$12 \times \cos 30^\circ$ is the horizontal run.

11. Answer D Concept 7 • transfer

A 10-ft ladder leans against a wall with its foot 5 ft from the wall. What angle does the ladder make with the ground?

Target word: angle, not side **Decides:** $\cos \theta = \text{adjacent/hypotenuse} = 5/10$.

A	Adjacent category	$\tan^{-1}(5/10)$ treats the ladder as a leg, not the hypotenuse.
B	Inverted operation	$\tan^{-1}(10/5)$ flips the ratio and misuses the hypotenuse.
C	Confusion set	$\sin^{-1}(0.5)$: treats 5 ft as the opposite side.
D	Correct	$\cos^{-1}(0.5) = 60^\circ$.

12. Answer B Concept 8

In a classroom ~~with new tablets and a geometry app the district bought last spring,~~ a student sorts shapes and says a figure cannot be a rectangle “because it looks like a square.” ~~The student is a strong reader and enjoys drawing.~~ Which next step best fits the student’s level of thinking?

Target word: task matched to Van Hiele level **Decides:** Judging by appearance is the visualization level; next is analysis of properties.

A	Camouflage	Answers an interest detail; drawing alone keeps the student at the visual level.
B	Correct	Moves a visual thinker toward property-based analysis.
C	Absolute language	Van Hiele levels come from experience, not age or grade.
D	Adjacent category	A deduction-level task that skips the analysis level.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Rigid motions vs. dilations	Translations, reflections, and rotations preserve length and angle (congruence); a dilation preserves angles but scales every length by k (similarity).	Calling any image made by a sequence of transformations congruent.	A scale factor appears anywhere in the sequence.
2	Triangle congruence criteria	SSS, SAS, ASA, AAS (and HL for right triangles) guarantee congruence; SSA and AAA do not.	Accepting any three matching parts, especially an angle that is not between the two sides.	Two sides and an angle: check whether the angle is included.
3	Similarity is multiplicative	In similar figures, corresponding sides share one ratio; missing lengths come from a proportion, not a constant difference.	Additive reasoning: adding the same amount to each side.	Shadows, scale drawings, or nested triangles with one unknown length.
4	Parallel and perpendicular slopes	Parallel lines have equal slopes; perpendicular lines have slopes that are negative reciprocals (product -1).	Only negating, or only flipping, the slope.	"Perpendicular to the line through..."
5	Volume of cones and pyramids	$V = (1/3)Bh$, using the perpendicular height; a cone holds one-third of the cylinder with the same base and height.	Dropping the $1/3$ or using the slant height.	A cone or pyramid with both a height and a slant height given.
6	Circle equation by completing the square	$(x - h)^2 + (y - k)^2 = r^2$ has center (h, k) and radius r ; complete the square in x and y to reach this form.	Flipping the signs of the center or reporting r^2 as the radius.	x^2 and y^2 with coefficient 1 plus linear x and y terms.
7	Right-triangle trigonometry	$\sin = \text{opposite/hypotenuse}$, $\cos = \text{adjacent/hypotenuse}$, $\tan = \text{opposite/adjacent}$, all measured from the given angle.	Naming sides from the wrong angle or picking the wrong ratio.	One angle and one side of a right triangle, one side wanted.
8	Van Hiele levels of geometric thought	Students move from visualization to analysis of properties to informal deduction to formal proof; tasks should meet the current level and push to the next one.	Jumping to formal proof, or assuming age or grade sets the level.	A student judges a shape by how it looks rather than its properties.

Course scope (public description): Geometry for Secondary Mathematics Teaching explores important conceptual underpinnings, common student misconceptions and ways of thinking, appropriate use of technology, and instructional practices to support and assess the learning of geometry. Students in this course will develop a deep understanding of constructions and transformations, congruence and similarity, analytic geometry, solid geometry, conics, trigonometry, and the historical development of content. Calculus I is a prerequisite for this course.